

REINFORCE++: STABILIZING CRITIC-FREE POLICY OPTIMIZATION WITH GLOBAL NORMALIZATION

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ABSTRACT

Reinforcement Learning from Human Feedback (RLHF) is used to align Large Language Models (LLMs). A widely used algorithm, Proximal Policy Optimization (PPO), employs a critic network to estimate advantages, which introduces significant computational and memory overhead. To address this, a family of critic-free algorithms (e.g., GRPO, RLOO) has emerged. However, methods such as GRPO rely on *prompt-level (local)* advantage normalization, whose finite-sample distortion we characterize and whose training stability and held-out generalization we examine. We introduce **REINFORCE++**, a critic-free framework centered on **Global Advantage Normalization**. By normalizing advantages across the entire global batch rather than small, prompt-specific groups, our method uses a shared scale estimated from the rollout batch. We characterize finite-sample self-normalization and its large-sample limit in an independent Gaussian reward model. We introduce two variants: **REINFORCE++**, a computationally efficient algorithm ($k \geq 1$) for general-domain RLHF, and **REINFORCE++_{w/ Baseline}**, a group-sampling variant ($k > 1$) for complex reasoning tasks. Experiments show competitive preference-alignment scores with single-response sampling and higher held-out reasoning scores than GRPO. On five tool-use benchmarks, REINFORCE++_{w/ Baseline} achieves a mean accuracy of 24.10, compared with 22.58 for GRPO and 21.85 for PPO.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) is a key technique for aligning Large Language Models (LLMs) with human values and preferences (Vemprala et al., 2023; Achiam et al., 2023; Ouyang et al., 2022; Shen & Zhang, 2024; Shen et al., 2025; Hu et al., 2024). Alongside preference-optimization methods such as DPO (Rafailov et al., 2023), systems including ChatGPT/GPT-4 (Vemprala et al., 2023; OpenAI, 2023), Claude (Anthropic, 2023), and Gemini (Team et al., 2023) have incorporated reinforcement learning into post-training.

A widely used RL algorithm in this setting is Proximal Policy Optimization (PPO) (Schulman et al., 2017). In actor-critic implementations of PPO, a learned value function provides a baseline for estimating advantages. However, this critic network introduces substantial computational overhead and memory requirements, increasing the resources required for model alignment (Shao et al., 2024).

To address PPO’s efficiency issues, a family of REINFORCE-based methods has emerged without the critic model, including ReMax (Li et al., 2023), RLOO (Ahmadian et al., 2024), and GRPO (Shao et al., 2024). These algorithms construct reward-based baselines using greedy decoding or multiple sampled responses to the same prompt.

These critic-free methods differ in their approach to *advantage estimation*. GRPO uses **prompt-level (local) normalization**, estimating a baseline and standard deviation from responses to one prompt; RLOO uses a leave-one-out baseline without inherently requiring this scale normalization. First, Appendix B shows that finite-sample self-normalization differs from population standardization. Second, small groups (e.g., $k = 4$ or $k = 8$) make the scale sensitive to sampled outcomes. Homogeneous groups provide no centered reward signal, while different reward spreads induce different prompt-specific scales. Prompt-specific scales also change the relative weighting of reward differences across prompts. Our experiments examine the resulting training dynamics and held-out generalization.

We introduce REINFORCE++, a framework with two variants centered on global advantage normalization. The basic variant supports $k \geq 1$. In its $k = 1$ configuration (detailed in Algorithm 1), it maximizes prompt diversity for general-domain RLHF. It can also be applied with $k > 1$ (Section 4.2), using the same global normalization principle. REINFORCE++_{w/ Baseline} is a specialized *variant* for complex tasks ($k > 1$) that benefits from **group sampling**. It separates reward centering from scale estimation by first subtracting the *group mean* (for within-prompt comparisons) and then normalizing by the *global standard deviation* (for a shared batch scale). It also employs a k_2 gradient surrogate for separate KL regularization, as detailed in Appendix C. In summary, our contributions are as follows:

- We provide a theoretical proof that finite-sample local normalization differs from population standardization under explicit sampling assumptions (Appendix B).
- We propose REINFORCE++, a critic-free framework centered on Global Advantage Normalization, to share normalization statistics across prompts while retaining PPO clipping.
- We present two algorithms: REINFORCE++ supports $k \geq 1$ for general RLHF and reasoning, while REINFORCE++_{w/ Baseline} combines group-mean reward centering with global normalization for $k > 1$.
- The experiments show that REINFORCE++ achieves competitive general RLHF scores with shorter responses and stronger held-out reasoning performance. The baseline variant achieves higher mean accuracy than GRPO and PPO across five tool-use benchmarks.

2 BACKGROUND AND RELATED WORK

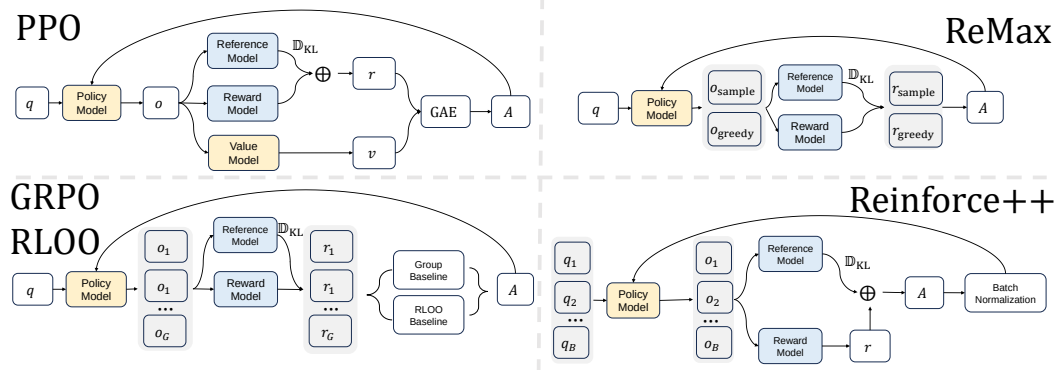


Figure 1: The comparison of PPO, ReMax, GRPO, RLOO, and REINFORCE++. REINFORCE++, as shown in the figure, removes the critic model and uses global batch normalization. REINFORCE++_{w/ Baseline} uses group sampling, similar to GRPO/RLOO, but combines group-mean subtraction with global batch normalization.

2.1 PPO AND CRITIC-FREE RLHF

PPO optimizes LLMs by maximizing the following surrogate objective:

$$\mathcal{L}_{\text{PPO}}(\theta) = \mathbb{E}_{q \sim P(Q), o \sim \pi_{\theta, \text{old}}(o|q)} \left[\frac{1}{|o|} \sum_{t=1}^{|o|} \min \left(s_t(\theta) A_t, \text{clip}(s_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right] \quad (1)$$

where $s_t(\theta) = \frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta, \text{old}}(o_t|q, o_{<t})}$ is the probability ratio. The advantage A_t is typically calculated using Generalized Advantage Estimation (GAE) (Schulman et al., 2015):

$$A_{q, o_t} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad (2)$$

where $\delta_{q,o_t} = r_t + \gamma V(o_{t+1}) - V(o_t)$ is the temporal difference error, and $V(\cdot)$ is the critic network. Critic-free methods in Figure 1 remove the critic $V(\cdot)$ and compute the advantage A_t directly from rewards.

- **ReMax** (Li et al., 2023) uses a greedy decoding response $\hat{o}$ as the baseline: $A_{q,o_t} = r(o) - r(\hat{o})$.
- **RLOO** (Ahmadian et al., 2024) samples k responses and uses the mean of others as the baseline: $A_{q,o_t^{(i)}} = r(o^{(i)}) - \frac{1}{k-1} \sum_{j \neq i} r(o^{(j)})$.
- **GRPO** (Shao et al., 2024) also samples k responses but normalizes the advantage using the mean and standard deviation of the *local group*:

$$A_{q,o_t^{(i)}} = \frac{r(o^{(i)}) - \text{mean}(\{r(o^{(j)})\}_{j=1}^k)}{\text{std}(\{r(o^{(j)})\}_{j=1}^k) + \epsilon} \quad (3)$$

2.2 PROPERTIES OF LOCAL NORMALIZATION

The **prompt-level (local) normalization** in Eq. (3) has three properties relevant to advantage estimation and evaluation:

1. **Finite-Sample Distortion:** Appendix B proves that self-normalization differs from the population-standardized reward in finite samples. The result concerns the conditional expectation of a self-normalized scalar reward under independent Gaussian sampling.
2. **Practical Sensitivity:** Small groups (e.g., 4 or 8 responses) provide noisy estimates of reward spread. Standardized advantages are bounded by $\sqrt{k-1}$ for nonconstant groups using divisor k . Identical rewards produce zero centered signal; mixed outcomes induce prompt-specific update scales.
3. **Prompt Weighting:** Group-relative updates compare responses to the same prompt and rescale each group by its own reward spread. Our small-data experiments evaluate training-set fit and held-out performance separately to examine generalization under this weighting.

3 METHOD

Our method uses normalization statistics pooled across the rollout batch for critic-free RLHF. We propose two variants for different use cases.

3.1 GLOBAL NORMALIZATION

The primary REINFORCE++ algorithm is designed for general-purpose RLHF. In its $k = 1$ configuration, it allocates each response in the rollout budget to a distinct prompt.

It optimizes the PPO objective in Eq. (1) but redefines the advantage A_{q,o_t} . We use the standard PPO reward formulation, which incorporates the kl-style KL penalty directly into the reward:

$$A_{q,o_t} = r(o_{1:T}, q) - \beta \cdot \sum_{i=t}^T \text{KL}(i) \quad (4)$$

where $\text{KL}(t) = \log \left(\frac{\pi_{\theta_{\text{old}}}(o_t | q, o_{<t})}{\pi_{\text{ref}}(o_t | q, o_{<t})} \right)$.

We use Global Advantage Normalization to share statistics across prompts, including the single-response setting where within-prompt reward variation cannot be estimated (Andrychowicz et al., 2020):

$$A_{q,o_t}^{\text{norm}} = \frac{A_{q,o_t} - \text{mean}(A | A \in \mathcal{D}_{\text{batch}})}{\text{std}(A | A \in \mathcal{D}_{\text{batch}}) + \epsilon} \quad (5)$$

The global normalization is the core of REINFORCE++. Pooling a large rollout batch (e.g., 1024 responses) replaces separate small-group scales with shared statistics. Appendix B characterizes convergence of the normalization statistics under independent sampling. Valid token positions exclude padding, and the moments remain fixed during rollout reuse. The efficient $k = 1$ case is detailed in Algorithm 1.

3.2 LOCAL BASELINE

For more complex tasks, such as multi-step reasoning, sampling multiple responses per prompt ($k > 1$) can be beneficial (Yue et al., 2025). To this end, we introduce REINFORCE++_{w/ Baseline}. This variant uses group sampling for reward comparisons and global normalization for scale estimation. The advantage calculation is a two-step process:

- **Group Mean Subtraction for Reshaping:** We first subtract the group mean reward. This baseline removes reward offsets shared by responses to the same prompt; the subsequent global normalization sets their common scale.

$$A'_{q,o_t^{(i)}} = r(q, o^{(i)}) - \frac{1}{k} \sum_{j=1}^k r(q, o^{(j)}) \quad (6)$$

- **Global Batch Normalization for Scaling:** We then normalize this initial advantage using the **global batch statistics** shared across prompts.

$$A_{q,o_t}^{\text{norm}} = \frac{A'_{q,o_t} - \text{mean}_{\text{batch}}(A')}{\text{std}_{\text{batch}}(A') + \epsilon} \quad (7)$$

This separates reward centering from scale estimation; the inclusive group baseline equals $(k - 1)/k$ times RLOO (Appendix D).

Furthermore, this variant employs a separate k_2 KL regularizer. At a fixed context with on-policy actions held fixed during differentiation, its expected pointwise gradient equals the reverse-KL gradient (Liu et al., 2025a). During old-policy rollout reuse, the same term serves as an off-policy surrogate (Appendix C). The maximized objective is:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}}(A^{\text{norm}}) - \lambda \mathbb{E}_{o \sim \pi_{\text{old}}} \left[\frac{1}{|o|} \sum_t \frac{1}{2} \left(\log \frac{\pi_{\theta}(o_t | q, o_{<t})}{\pi_{\text{ref}}(o_t | q, o_{<t})} \right)^2 \right]. \quad (8)$$

3.3 RELATIONSHIP WITH PPO

Both variants retain PPO clipping without a learned critic. Setting $V = 0$ and $\gamma = \lambda_{\text{GAE}} = 1$ reduces GAE to Monte Carlo returns, the structure used by the basic variant before normalization. REINFORCE++_{w/ Baseline} additionally uses prompt-specific reward centering and a separate KL loss; these changes distinguish it from standard actor-critic PPO.

3.4 IMPLEMENTATION DETAILS

Token-level Advantage The task’s immediate reward is zero before termination, but its undiscounted return contributes to every preceding response token in Eq. (4). REINFORCE++_{w/ Baseline} broadcasts its centered terminal reward to valid response positions. The objective averages token losses within each response and then across responses, as specified in Appendix A.

Batch Construction For REINFORCE++ ($k = 1$), a global batch of size N comprises N distinct prompts. For REINFORCE++_{w/ Baseline} ($k = 4$), a global batch of size $N = 1024$ might contain $1024/4 = 256$ unique prompts. The global moments pool valid response positions across the rollout batch.

Mini-Batch Updates We partition each rollout batch into mini-batches for policy optimization:

- **Batch Processing:** Each update evaluates the clipped objective on a subset of the rollout batch.
- **Multiple Updates:** Reuse detached advantages and old-policy probabilities over E optimization epochs.
- **Stochastic Optimization:** Mini-batch sampling determines which responses contribute to each update.

Reward Normalization and Clipping Reward preprocessing controls the location, range, and scale of reward values:

- **Normalization:** Centers rewards and divides by a shared batch standard deviation.
- **Clipping:** Bounds reward values before advantage computation.
- **Scaling:** Sets the magnitude of the task reward relative to the KL penalty.

3.5 STATISTICAL MOTIVATION

Omitting the numerical stabilizer, under independent sampling with nonzero population variance, normalization approaches the population-standardized target as $N \rightarrow \infty$ (Appendix B). This motivates global normalization in both REINFORCE++ and REINFORCE++_{w/ Baseline}: a batch of 1024 responses pools more observations than groups of 4 or 8. Token dependence and repeated responses limit the effective independent sample count. The scalar convergence analysis describes scale estimation; clipping and rollout reuse enter separately through the optimization objective.

3.6 SUMMARY

The two variants share **Global Advantage Normalization** and a critic-free clipped objective. Instead of normalizing within a prompt’s local group, REINFORCE++ normalizes the advantage function across the *entire global training batch*. The overall approach is:

- **Shared Statistics:** Sample moments converge under independent sampling. For token-level statistics, within-trajectory dependence determines the effective sample size (Appendix B).
- **Critic-Free Optimization:** Both variants avoid a learned value network and its associated forward passes, gradient updates, and optimizer state.
- **Held-Out Evaluation:** Using a shared scale is associated with improved held-out reasoning results in our evaluated settings.

Algorithm 1 REINFORCE++ when $k = 1$

Require: Initial policy model π_{ref} , reward models R , task prompts $\mathcal{D}$

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1: policy model  $\pi_\theta \leftarrow \pi_{ref}$ 
2: for step = 1, ...,  $M$  do
3:   Sample a batch  $\mathcal{D}_{batch}$  from  $\mathcal{D}$ 
4:   Update the old policy model  $\pi_{old} \leftarrow \pi_\theta$ 
5:   Sample output ( $k = 1$ )  $o \sim \pi_{old}(\cdot | q)$  for each question  $q \in \mathcal{D}_{batch}$ 
6:   Compute rewards  $r_i$  for each sampled  $o_i$  using  $R$ 
7:   Compute advantage  $A_{q,o_t}$  via Eq. (4)
8:   Normalize advantages globally across  $\mathcal{D}_{batch}$  to get  $A_{q,o_t}^{norm}$  via Eq. (5)
9:   for iteration = 1, ...,  $E$  do
10:    Update  $\pi_\theta$  by maximizing the objective in Eq. (1) using  $A_{q,o_t}^{norm}$ 
11:   end for
12: end for
```

Ensure: π_θ

4 EXPERIMENTS

We evaluate our two algorithms, REINFORCE++ and REINFORCE++_{w/ Baseline}, in their respective target domains using the OpenRLHF framework (Hu et al., 2024).

4.1 GENERAL RLHF

First, we evaluate REINFORCE++ in its $k = 1$ configuration, focusing on general-domain RLHF, where efficiency and prompt diversity are key.

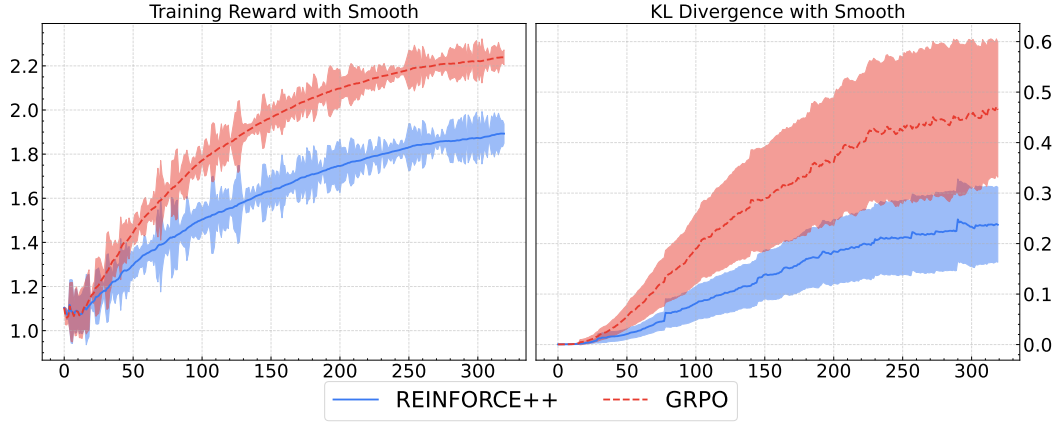


Figure 2: Smoothed training reward and KL divergence. REINFORCE++ ($k = 1$) shows a more gradual reward increase and lower KL divergence than GRPO over the plotted training interval.

Table 1: Comparison on Chat-Arena-Hard (Li et al., 2024). The single-sample REINFORCE++ ($k = 1$) scores 46.7 compared with 46.8 for GRPO, with shorter mean responses. Score/length divides the evaluation score by mean response length.

	Score	Length	Score/length
REINFORCE++ ($k=1$)	46.7	832	0.0561
GRPO ($k=4$)	46.8	860	0.0544
RLOO ($k=4$)	44.6	866	0.0515
ReMax ($k=1+1$)	45.1	805	0.0560

Experimental Setup We use an instruction-following policy model (Llama-3-8B-SFT) and refine it using a Bradley-Terry reward model (Bai et al., 2022) trained on $\sim 700K$ human preference pairs. The policy is trained on 20,000 diverse prompts. We compare REINFORCE++ ($k = 1$) with other critic-free methods: GRPO ($k = 4$), RLOO ($k = 4$), and ReMax ($k = 1 + 1$).

Experimental Results As shown in Table 1, the single-sample REINFORCE++ ($k = 1$) achieves a score of 46.7, close to the group-sampling GRPO (46.8). However, it produces shorter responses (832 tokens vs. 860), yielding a higher score/length ratio (0.0561). These measurements describe evaluation performance and generated response length for the single-response and group-sampling configurations.

Results Analysis Figure 2 shows the training dynamics. GRPO’s reward and KL divergence rise rapidly, while REINFORCE++ exhibits a more gradual reward increase and smaller divergence. These curves describe the reward–KL trade-offs of the compared training configurations. Response length can also influence preference-based evaluation (Dubois et al., 2024). We report evaluation scores alongside response lengths to make both aspects of the comparison explicit.

4.2 REASONING EXPERIMENTS

Next, we evaluate REINFORCE++ (group-sample, $k > 1$) in its target domain: complex reasoning, where group sampling is standard, and reward signals are often sparse (e.g., 0/1). We compare it directly with GRPO ($k > 1$).

4.2.1 LONG REASONING TASK

We use rule-based rewards (RLVR) in mathematical reasoning settings (Guo et al., 2025; Seed et al., 2025).

Table 2: Comparison on a small training dataset. Training on AIME-24 and evaluation on AIME-25. REINFORCE++ has lower training accuracy and higher held-out scores than GRPO.

Pass@N	AIME-24 (Train)	AIME-25 (Test)	
	N = 1	N = 1	N = 16
GRPO	95.0	0.0	0.4
REINFORCE++	71.0	2.5	40.0

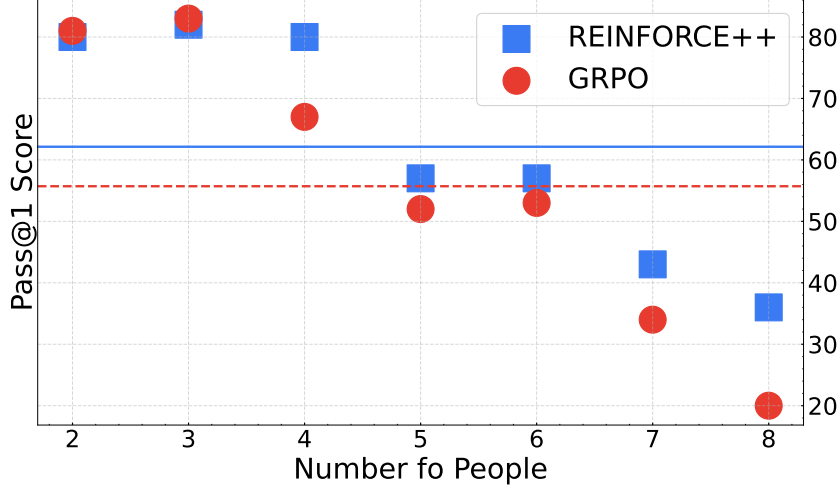


Figure 4: Comparison on logic benchmarks. REINFORCE++ achieves higher accuracy than GRPO on puzzles with four or more characters, including the eight-character evaluation.

Analysis on Small-Scale Datasets To test for overfitting, we trained on only 30 questions from AIME-24 and evaluated on AIME-25. As shown in Table 2, GRPO (local norm) achieves a near-perfect 95.0% on the training set but scores only 0.0 Pass@1 and 0.4 Pass@16 on the test set, showing a large train–test performance gap. In contrast, REINFORCE++ with global norm, while scoring lower on the training set (71.0), shows higher held-out scores (2.5 Pass@1, 40.0 Pass@16).

The training curves in Figure 3 show faster fitting of the training questions by GRPO and a more gradual increase for REINFORCE++.

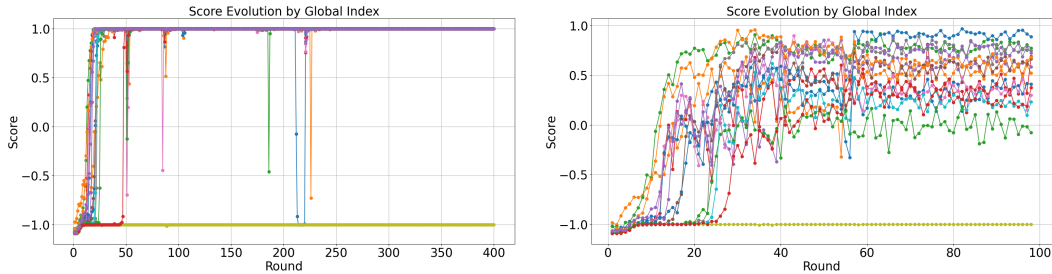


Figure 3: Training curves on a small prompt dataset. Left: GRPO with local normalization. Right: REINFORCE++ with global normalization. The curves show training dynamics; held-out scores appear in Table 2.

Logical Reasoning (K&K Puzzles) We also tested on the Knights and Knaves (K&K) puzzles (Xie et al., 2025; 2024), where difficulty increases with the number of characters in each puzzle. Figure 4 shows that while GRPO is competitive on easy tasks (2-3 people), it has lower accuracy on the harder eight-character puzzles. REINFORCE++ outperforms GRPO on the evaluated tasks with four or more characters and achieves a higher average score (62.1 vs. 55.7).

Table 3: Comparison on RL from Zero. REINFORCE++ achieves higher Pass@8 on AIME-24 and AMC-23; GRPO achieves higher Pass@1 on MATH-500.

Pass@N	AIME-24 (OOD) N = 8	AMC-23 (OOD) N = 8	MATH-500 (ID) N = 1
GRPO	18.96	59.22	73.00
REINFORCE++	21.04	60.47	72.00

Table 4: Tool-use benchmark accuracy (average@32). REINFORCE++_{w/ Baseline} achieves the highest five-benchmark mean and leads on four benchmarks.

Algorithm	AIME 24	AIME 25	HMMT 2025	HMMT 2024	CMIMC	Avg
GRPO (local norm)	31.66	21.87	16.97	17.70	24.68	22.58
PPO (critic-based)	30.20	21.66	15.00	18.43	23.95	21.85
RF++-Baseline (global norm)	30.83	27.18	17.91	18.95	25.62	24.10

RL from Zero Setting Finally, we applied reinforcement learning directly to a Qwen2.5-Math-Base model on MATH dataset splits (Guo et al., 2025). Table 3 shows that REINFORCE++ again achieves better OOD generalization on the more challenging AIME-24 and AMC-23 datasets, while remaining competitive on the in-distribution MATH-500 test set.

4.3 MULTI-STEP REINFORCEMENT LEARNING

Finally, to test performance in a complex, multi-step environment, we use the zero-shot agent setup from Mai et al. (2025), in which a Qwen 2.5 Base 7B model must learn to use Python tools to solve mathematical problems. We evaluate REINFORCE++_{w/ Baseline} with group sampling ($k > 1$) and group-mean reward centering in this setting (see Section 5).

Experimental Setup We adhere to the training and evaluation protocols of the ZeroTIR environment Mai et al. (2025). The backbone model is Qwen 2.5 Base 7B, trained using OpenRLHF Hu et al. (2024) with datasets from ORZ Hu et al. (2025) and DAPO Yu et al. (2025). We assess performance on benchmarks including AIME 2024 Jia (2024), AIME-2025 math ai (2025), HMMT FEB-2024/2025, and CMIMC, using the average@32 metric.

Experimental Result As shown in Table 4, REINFORCE++_{w/ Baseline} achieves the highest average accuracy (24.10) across all benchmarks. It outperforms both GRPO (22.58) and the full-critic PPO (21.85). The improvements in mean accuracy are 1.52 and 2.25 percentage points, respectively. The baseline variant leads on four of the five benchmarks, while GRPO leads on AIME-24. These results evaluate the combined use of reward centering, global normalization, and separate KL regularization.

5 PRACTICAL CONSIDERATIONS

5.1 SAMPLING AND REWARD DESIGN

The two variants differ in how they allocate a response budget and construct the reward baseline. With B responses per rollout batch, single-response sampling can cover B prompts, whereas sampling k responses per prompt covers B/k prompts. The latter allocation provides within-prompt reward comparisons for the baseline variant.

For multi-turn tool use, group-mean centering compares each response with alternatives to the same prompt. A group with identical rewards has zero centered task reward before batch normalization, while a mixed-outcome group retains positive and negative reward differences. This operation changes the reward signal without removing responses from the training batch. Subtracting the group mean also removes any reward offset common to the group. For example, changing binary rewards from 0/1 to $-1/1$ multiplies the centered rewards by two; subsequent global standardization cancels

this factor when the numerical stabilizer is omitted. With token-weighted global moments, the batch mean of the centered rewards can be nonzero when response lengths differ.

For general-domain RLHF, the $k = 1$ configuration allocates the rollout budget to broader prompt coverage. Section 4.1 reports competitive preference-alignment scores and shorter responses than GRPO under this configuration. For tasks with multiple sampled responses, the baseline variant provides prompt-specific comparisons together with a scale shared across the batch. Reward design, response length, and the relative weight of KL regularization jointly determine the resulting training signal.

5.2 EVIDENCE FROM RELATED SYSTEMS

Several subsequent studies examine global normalization in reasoning systems. Their experiments provide comparisons across reward schemes, model scales, and output-length constraints.

LitePPO (Liu et al., 2025c) combines group-mean centering, batch-level standard deviation, and token-level loss aggregation. Its normalization experiments favor the global standard deviation when paired with a group-level mean, while comparisons of full batch normalization vary with the reward scheme. These results support treating reward centering and scale estimation as separate design choices.

ScaleRL (Khatri et al., 2025) compares prompt-level, batch-level, and no variance normalization, reporting similar performance with a marginal advantage for batch-level normalization. Its larger-scale leave-one-out study includes runs with a 16,000 GPU-hour budget. This provides evidence for using shared batch statistics within a larger training recipe.

DLER (Liu et al., 2025b) evaluates normalization under output truncation. In its DeepSeek-R1-7B experiment, batch-wise normalization preserves AIME-24 accuracy as response length decreases, whereas group-wise normalization shows declining accuracy. This comparison identifies output-length control as another setting in which the choice of normalization affects training behavior.

6 CONCLUSION

We present REINFORCE++, a critic-free framework that combines PPO clipping with global advantage normalization. The framework separates the choice of reward baseline from the scale used across prompts. Our scalar analysis characterizes finite-sample self-normalization and its convergence to population standardization under independent Gaussian sampling with a vanishing numerical stabilizer.

The two variants implement this framework for different sampling configurations:

1. REINFORCE++ uses globally normalized returns with a KL penalty incorporated into the reward. The $k = 1$ configuration provides one response per prompt, while $k > 1$ supports multiple sampled responses in reasoning tasks.
2. REINFORCE++_{w/ Baseline} combines group-mean reward centering, global normalization, and a separate k_2 KL surrogate. It uses within-prompt reward comparisons while sharing normalization statistics across the rollout batch.

Experiments cover preference alignment, mathematical and logical reasoning, and multi-step tool use. The single-response variant scores 46.7 on Chat-Arena-Hard, compared with 46.8 for GRPO, with shorter mean responses. The reasoning experiments show higher held-out scores than GRPO across the reported AIME and AMC comparisons. In tool use, the baseline variant achieves the highest mean accuracy across five benchmarks, exceeding GRPO and PPO by 1.52 and 2.25 percentage points. Together, these results support global advantage normalization as a practical component of critic-free policy optimization across the evaluated tasks.

AI USE STATEMENT

Generative AI tools assisted with manuscript preparation, including formatting, mathematical exposition, and proof revision. Experimental data and results were produced by the authors. The authors reviewed the AI-assisted content and take responsibility for the final manuscript and its accompanying artifacts.

ETHICS STATEMENT

Reinforcement learning is increasingly used to steer large language models by optimizing them against a reward signal, including verifiable, programmatic rewards (RLVR) for domains such as math, code, and tool use. The main potential positive impact of this line of work is improved scalability and reproducibility of post-training: verifiable rewards can reduce reliance on human labeling and make optimization targets easier to audit and replicate. The critic-free methods studied here aim to support this use of reinforcement learning through shared advantage scaling and prompt-specific reward centering.

Potential risks include reward gaming when verifiers are incomplete or mis-specified, and dual-use concerns if improved tool-use or problem-solving capabilities are deployed without safeguards. We recommend evaluating beyond verifier scores (e.g., robustness and red-teaming), stress-testing verifiers for exploitability, and using conservative deployment controls (e.g., access restrictions and monitoring) for systems trained with RL.

REPRODUCIBILITY STATEMENT

Section 3 specifies the algorithms, and Section 4 describes the experimental settings. Appendix A specifies the response-wise loss reduction of the policy objective. Appendices B and C provide assumptions and derivations for normalization and KL regularization.

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A RESPONSE-WISE LOSS REDUCTION AND IMPLEMENTATION

The objective in Eq. (1) uses a mean over valid tokens within each response, followed by a mean over responses in the batch. This section specifies the reduction independently of advantage normalization.

Let $m_{i,t} \in \{0, 1\}$ indicate a valid response token included in the policy loss, and let $n_i = \sum_t m_{i,t} > 0$. For B responses, the policy loss is

$$L_{\text{PG}}(\theta) = \frac{1}{B} \sum_{i=1}^B \left[\frac{1}{n_i} \sum_t m_{i,t} \ell_{i,t}(\theta) \right], \quad \ell_{i,t}(\theta) = -f_{i,t}(\theta, \text{sg}[\hat{A}_{i,t}]). \quad (9)$$

Here $\hat{A}$ denotes A^{norm} , f is the clipped surrogate in Eq. (1), and sg denotes stop-gradient. Padding and positions outside the response loss mask do not enter the numerator or token count. The separate k_2 term in Eq. (8) uses the same reduction.

Response weighting. Each response has outer weight $1/B$, and each valid token within response i has weight $1/(Bn_i)$. This differs from dividing the sum of all token losses by the total number of valid tokens:

$$L_{\text{token}} = \frac{\sum_{i,t} m_{i,t} \ell_{i,t}}{\sum_i n_i}. \quad (10)$$

The latter weights each response in proportion to its length. The two reductions agree when all responses have equal length. With a fixed number k of responses per prompt, batch averaging is also equivalent to averaging the per-response losses within each prompt and then across prompts.

Minibatches and accumulation. For a minibatch $\mathcal{M}$, replace B by $|\mathcal{M}|$ in Eq. (9). To recover the same mean when combining uneven microbatches or distributed shards, their loss sums must be weighted by their response counts. The normalization moments are associated with the rollout batch; changing the partition into minibatches does not change the definition of the already computed advantages.

Terminal reward and token returns. For the basic variant, write $d_{i,t} = \log(\pi_{\text{old}}/\pi_{\text{ref}})$ at the sampled token. The immediate reward can be written as

$$\tilde{r}_{i,t} = \mathbf{1}\{t = T_i\}r(q_i, o_i) - \beta d_{i,t}. \quad (11)$$

With unit discount, its return is $\sum_{u=t}^{T_i} \tilde{r}_{i,u} = A_{q_i, o_i, t}$ from Eq. (4). Thus, a terminal task reward contributes to the return at every preceding response position. For REINFORCE++_{w/ Baseline}, the group-centered terminal reward in Eq. (6) is broadcast to response positions before global normalization.

B FINITE-SAMPLE PROPERTIES OF NORMALIZATION

We analyze a scalar model to make the effect of estimating normalization statistics from the same samples explicit. Let

$$r_i = \mu + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma > 0, \quad (12)$$

and define $\bar{\epsilon} = N^{-1} \sum_i \epsilon_i$, $D_N^2 = N^{-1} \sum_i (\epsilon_i - \bar{\epsilon})^2$, and $Z_{i,N} = (\epsilon_i - \bar{\epsilon})/D_N$ for $N \geq 2$. The population-standardized target is ϵ_i/σ . We omit the numerical stabilizer for this analysis; $D_N > 0$ almost surely under the stated model.

Proposition B.1 (Finite-sample bound). *For any nonconstant sample of size $N \geq 2$, $|Z_{i,N}| \leq \sqrt{N-1}$. Consequently, under the Gaussian model, $\mathbb{E}[Z_{i,N} \mid \epsilon_i]$ cannot equal ϵ_i/σ almost surely for finite N .*

Proof. Set $x_j = \epsilon_j - \bar{\epsilon}$, so $\sum_j x_j = 0$. By Cauchy–Schwarz,

$$x_i^2 = \left(\sum_{j \neq i} x_j \right)^2 \leq (N-1) \sum_{j \neq i} x_j^2. \quad (13)$$

Therefore $\sum_j x_j^2 \geq Nx_i^2/(N-1)$, giving $Z_{i,N}^2 = Nx_i^2/\sum_j x_j^2 \leq N-1$. The conditional expectation obeys the same bound, whereas $|\epsilon_i|/\sigma > \sqrt{N-1}$ has positive probability for a Gaussian variable. Equality with the standardized target cannot hold almost surely. $\square$

Conditional moments of the numerator and denominator. The centered numerator satisfies

$$\mathbb{E}[\epsilon_i - \bar{\epsilon} \mid \epsilon_i] = \left(1 - \frac{1}{N}\right) \epsilon_i. \quad (14)$$

Using $D_N^2 = N^{-1} \sum_j \epsilon_j^2 - \bar{\epsilon}^2$ and independence,

$$\mathbb{E}[\bar{\epsilon}^2 \mid \epsilon_i] = \frac{\epsilon_i^2 + (N-1)\sigma^2}{N^2}, \quad (15)$$

$$\mathbb{E}[D_N^2 \mid \epsilon_i] = \frac{(N-1)^2}{N^2} \sigma^2 + \frac{N-1}{N^2} \epsilon_i^2. \quad (16)$$

These identities describe how the included observation affects the normalization statistics. Proposition B.1 establishes the finite-sample discrepancy through a direct bound on the ratio.

An exact two-response example. For $N=2$, $D_2 = |\epsilon_1 - \epsilon_2|/2$ and $Z_{1,2} = \text{sign}(\epsilon_1 - \epsilon_2)$ almost surely. Thus,

$$\mathbb{E}[Z_{1,2} \mid \epsilon_1 = x] = 2\Phi(x/\sigma) - 1, \quad (17)$$

where Φ is the standard normal cumulative distribution function. This bounded, nonlinear function differs from the linear population-standardized value x/σ . It illustrates how a small group changes the reward scale through self-normalization.

Proposition B.2 (Large-sample limit). *Under the Gaussian model, for any fixed i , $Z_{i,N} \rightarrow \epsilon_i/\sigma$ almost surely and in L^1 as $N \rightarrow \infty$. In particular,*

$$\mathbb{E}[|\mathbb{E}[Z_{i,N} \mid \epsilon_i] - \epsilon_i/\sigma|] \rightarrow 0. \quad (18)$$

Proof. The strong law gives $\bar{\epsilon} \rightarrow 0$ and $D_N^2 \rightarrow \sigma^2$ almost surely, establishing the first limit. By exchangeability and $\sum_{j=1}^N Z_{j,N}^2 = N$, we have $\mathbb{E}[Z_{i,N}^2] = 1$ for every N . This uniform second-moment bound implies uniform integrability, so the almost-sure convergence also holds in L^1 . Conditional Jensen's inequality gives

$$\mathbb{E}[|\mathbb{E}[Z_{i,N} - \epsilon_i/\sigma \mid \epsilon_i]|] \leq \mathbb{E}[|Z_{i,N} - \epsilon_i/\sigma|] \rightarrow 0. \quad (19)$$

$\square$

Numerical stabilizer. With a fixed stabilizer $\eta > 0$, $(\epsilon_i - \bar{\epsilon})/(D_N + \eta)$ converges instead to $\epsilon_i/(\sigma + \eta)$. The unstabilized population-standardization limit applies when the stabilizer is omitted or tends to zero. The finite-sample bound remains valid with a nonnegative stabilizer for nonconstant samples; constant samples have zero numerator and return zero when the stabilizer is positive.

Implications for global normalization. The propositions isolate the effect of normalization sample size: estimated moments approach their population values as the number of independent observations grows. A shared batch scale can use observations across many prompt groups, rather than estimating a separate scale from each small group. For grouped or token-level data, the interpretation depends on the number of independently sampled trajectories and their dependence structure.

Relationship to baselines and gradients. For an unscaled group mean, $\mathbb{E}[r_i - \bar{r}_q \mid r_i, q] = (1 - 1/k)(r_i - \mu_q)$ when responses are conditionally independent with mean reward μ_q . This is the same factor appearing in Eq. (24). The propositions describe convergence to a standardized scalar target. Policy-gradient expectations additionally involve the joint distribution of these normalized rewards and the policy score function.

C KL REGULARIZATION AS A GRADIENT SURROGATE

For a fixed context, write $p_\theta(y) = \pi_\theta(y \mid h)$, $q(y) = \pi_{\text{ref}}(y \mid h)$, and $z(y) = \log[p_\theta(y)/q(y)]$. Assume common support and sufficient regularity to interchange differentiation and expectation. The reverse KL is

$$D_{\text{KL}}(p_\theta \| q) = \mathbb{E}_{y \sim p_\theta}[z(y)], \quad \nabla_\theta D_{\text{KL}}(p_\theta \| q) = \mathbb{E}_{p_\theta}[z(y) \nabla_\theta \log p_\theta(y)], \quad (20)$$

using $\mathbb{E}_{p_\theta}[\nabla_\theta \log p_\theta] = 0$.

Value estimators and differentiated losses. The common expressions are

$$k_1(z) = z, \quad k_2(z) = \frac{1}{2}z^2, \quad k_3(z) = e^{-z} - 1 + z. \quad (21)$$

Both k_1 and k_3 have expectation $D_{\text{KL}}(p_\theta \| q)$ under on-policy sampling, since $\mathbb{E}_{p_\theta}[e^{-z}] = 1$. When sampled actions are treated as fixed during automatic differentiation, their pointwise gradients differ:

$$\begin{aligned} \nabla_\theta k_1 &= \nabla_\theta \log p_\theta, \\ \nabla_\theta k_2 &= z \nabla_\theta \log p_\theta, \\ \nabla_\theta k_3 &= (1 - q/p_\theta) \nabla_\theta \log p_\theta. \end{aligned} \quad (22)$$

The on-policy expectation of the k_2 pointwise gradient is exactly Eq. (20). This is the gradient-surrogate interpretation used for the separate KL term in REINFORCE++_{w/ Baseline} (Liu et al., 2025a). It is distinct from differentiating the entire expectation $\mathbb{E}_{p_\theta}[k_2]$, which also differentiates the sampling distribution.

Under the same fixed-context assumptions, the expected pointwise gradient of k_1 is zero. The k_1 expression can instead enter the reward and act as a detached score-function coefficient, as in the returns used by the basic variant. The expected pointwise gradient of k_3 equals $-\mathbb{E}_q[\nabla_\theta \log p_\theta]$, the forward-KL gradient. Its value-estimation and loss-gradient roles should therefore be distinguished.

Numerical sensitivity. The k_3 pointwise gradient contains the importance ratio q/p_θ , which can become large when the current policy assigns low probability to actions favored by the reference. Computing this ratio as $\exp(-z)$ can also overflow for sufficiently negative z . The k_2 surrogate uses the log ratio in place of this exponential ratio. Its gradient magnitude and variance depend on the joint distribution of the log ratio and the score function.

Rollout reuse. The on-policy identities apply when the sampled actions follow the current policy. For actions sampled from p_{old} , a fixed-context importance-weighted gradient satisfies

$$\mathbb{E}_{p_{\text{old}}} \left[\frac{p_\theta(y)}{p_{\text{old}}(y)} z(y) \nabla_\theta \log p_\theta(y) \right] = \nabla_\theta D_{\text{KL}}(p_\theta \| q). \quad (23)$$

The practical separate loss in Eq. (8) is the unweighted k_2 surrogate on the rollout data. Its exact gradient-equivalence interpretation applies at the on-policy point; subsequent optimization uses it as a surrogate. For autoregressive policies, the fixed-context statement concerns the conditional next-token distribution.

D DESIGN IMPLICATIONS AND SAMPLING BUDGETS

Local baseline and global scale. For a prompt with $k > 1$ responses, let $C_{q,i} = r_i - k^{-1} \sum_j r_j$. Then

$$C_{q,i} = \frac{k-1}{k} \left(r_i - \frac{1}{k-1} \sum_{j \neq i} r_j \right) = \frac{k-1}{k} A_i^{\text{RLOO}}. \quad (24)$$

The group baseline compares a response with alternatives to the same prompt; global scaling supplies a shared scale across prompts. Homogeneous groups have zero centered task-reward signal. This separates reward centering from prompt-specific standard-deviation normalization.

Response budget. At fixed response count B , single-response sampling covers up to B prompts; grouped sampling covers B/k . Larger groups trade prompt coverage for exploration of each prompt. Runtime and memory also depend on response length, batching, optimizer state, and rollout implementation.

Evaluation and attribution. The tool-use metric average@32 averages success across 32 sampled attempts; it differs from pass@32, which counts success if at least one attempt succeeds. The reported tool-use mean is an unweighted average across five benchmarks. The improvements of 1.52 and 2.25 percentage points over GRPO and PPO concern the combined training recipe. Component-level attribution requires controlled comparisons of normalization, reward centering, KL regularization, and loss reduction.